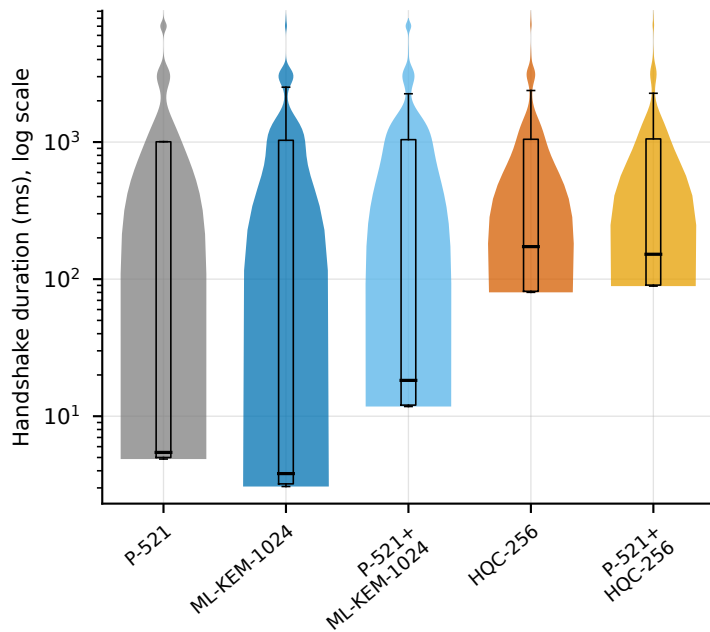


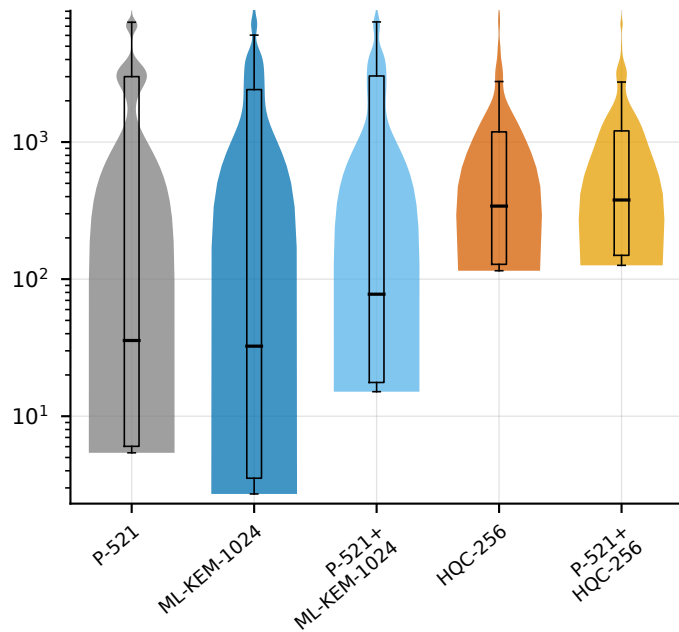
(a)

secp521r1 (classical)



(b)

mldsa87 (post-quantum)



(c)

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	secp521r1 Fail	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)	mldsa87 Fail
P-521	910.7	196.2	10.32k	26/500	1.43k	155.6	9.07k	44/500
ML-KEM-1024	775.5	180.4	11.17k	10/500	1.60k	165.4	16.53k	23/500
P-521+ML-KEM-1024	912.9	174.4	8.02k	7/500	1.56k	150.0	11.19k	26/500
HQC-256	751.1	175.6	10.15k	1/500	1.07k	163.9	17.87k	9/500
P-521+HQC-256	756.6	193.2	15.84k	4/500	1.04k	145.2	14.60k	5/500